

Structural refinements and formal certificates for C_7 and C_{13} zero-error codes

4 October 2026

Abstract

We study simultaneous mask allocation on three actual fibers of a strong power of the thirteen-cycle. For a center of prescribed size a and two leaves of required sizes at least b, c , the pairwise budgets $a + b, a + c \leq 169$ do not characterize feasibility. In the small-slack regime $u = 169 - a - b < 13$, $v = 169 - a - c < 13$, the exact additional condition is $\min(a, 169 - a) \leq uv$. A minority-rectangle argument proves necessity for arbitrary masks; a Ferrers diagram or its complement proves sufficiency. The result is realized in $C_{13}^{\boxtimes 6}$ and formalized in Lean, including the sharp balanced occupancy 80 and a 240-point example. For context we publish complete formal certificates for the existing bounds $\Theta(C_7) > 3.258834805519757$ and $\Theta(C_{13}) > 6.302927403571109$, obtained within inherited recursive and guarded-cell methods. An appendix gives a classical correlated C_7^3 illustration with full coordinate margins. The local theorem and that illustration do not improve these capacity bounds.

1 Scope, notation and construction context

Let C_q be the cycle on $\mathbb{Z}/q\mathbb{Z}$. Its closed confusability relation is $x \simeq y$ when $x - y \in \{0, 1, -1\}$. On a strong power, two words are confusable exactly when they are confusable in every coordinate. A zero-error code is a finite set in which distinct words are not confusable, equivalently an independent set of the strong power. We use

$$\Theta(G) = \sup_{d \geq 1} \alpha(G^{\boxtimes d})^{1/d}.$$

For sets S, T write $S \perp T$ if no point of S is confusable with a point of T . This includes exclusion of equality. Write $\mathcal{N}(S)$ for the closed neighborhood of S .

The main question here is structural: how can two different directional separation requirements constrain a *shared* center? Our answer is an exact local region, not a new general coding framework. Its elementary proof is self-contained. Ferrers compression and discrete isoperimetry are classical; grid and torus inequalities of Bollobás–Leader provide relevant context [8, 9], but are not invoked as a theorem implying our allocation formula. We do not assert worldwide priority for the elementary ingredients.

The capacity constructions placed alongside this result have a different role. Gao isolated a recursive private-pair invariant [1]; Buys–Polak–Zuiddam (BPZ) supplied improved finite data, layered substitutions and Lean certificates [2, 3]; Tandon enlarged the auxiliary-set construction by one-sided heterogeneity [4]. Protti introduced guarded typed-cell refinements [5], and Long published improved recursion DAGs [6]. The retained C_7 certificate uses the Gao–BPZ–Tandon methods. The retained C_{13} certificate is a new concrete combination of Protti’s guarded cells with Long’s DAG, not the invention of guarded cells. The local mask theorem below addresses a question not answered by scalar pairwise size budgets in these constructions.

Table 1 compares *chosen construction roots*, using Stavriianov’s C_7 certificate [7] and Long’s C_{13} certificate [6]. These are not the unknown capacities and no global optimality is

asserted. All source versions in the comparison are fixed, as identified in the references and the companion’s `NOTICE.md`.

Cycle	Selected prior construction	Retained construction	Dimensions
C_7	Stavriianov: 3.258834362237710794...	3.258834805519757...	560 / 300
C_{13}	Long: 6.302927071589786110...	6.302927403571109071...	522 / 522

Table 1: Prior and retained code roots, not equalities for $\Theta(C_q)$. The structural round reported in this paper adds no capacity gain.

The companion repository contains the paper, mathematical construction data, and a portable, pinned Lean project. The single aggregate module `StructuralRelease` exports aliases for both strict capacity endpoints, the actual local region, the physical example and the C_7 illustration. The argument proceeds from a combinatorial lemma to the exact region, then to its graph realization; Sections 5–6 give the contextual construction and certificate interfaces.

2 A minority rectangle on a cyclic grid

Consider a binary matrix $f : X \times Y \rightarrow \{0, 1\}$ on finite sets. A row $\{x\} \times Y$ is *mixed* if both colors occur; a column $X \times \{y\}$ is mixed in the analogous sense. Denote their index sets by $R_f \subseteq X$ and $Q_f \subseteq Y$.

Lemma 2.1 (Minority rectangle). *If $|R_f| < |X|$ and $|Q_f| < |Y|$, one color occurs only in $R_f \times Q_f$. In particular,*

$$\min(|f^{-1}(1)|, |f^{-1}(0)|) \leq |R_f| |Q_f|.$$

Proof. Choose an unmixed row and an unmixed column. Their constant colors agree at the intersection; call this color ε . Every unmixed row meets the chosen unmixed column, so has color ε , and every unmixed column meets the chosen unmixed row, so has that same color. A point of the other color must therefore lie on both a mixed row and a mixed column. Counting their intersections proves the inequality. Notice that the color singled out by the argument need not be specified in advance. $\square$

Set $H = (\mathbb{Z}/13\mathbb{Z})^2$, $e_1 = (1, 0)$ and $e_2 = (0, 1)$. For $A \subseteq H$ define its two directional exit boundaries and their sizes by

$$\partial_i A = (A + e_i) \setminus A, \quad d_i(A) = |\partial_i A| \quad (i = 1, 2).$$

Thus $|A \cup (A + e_i)| = |A| + d_i(A)$. A mixed column, with the second coordinate fixed, has at least one transition from membership to nonmembership while cycling in direction e_1 . These transitions occur in distinct columns; consequently $|Q_A| \leq d_1(A)$. Similarly $|R_A| \leq d_2(A)$.

Proposition 2.2 (Directional minority bound). *For any $A \subseteq H$ with $d_1(A), d_2(A) < 13$,*

$$\min(|A|, 169 - |A|) \leq d_1(A) d_2(A). \tag{1}$$

Proof. The bounds on the mixed-line counts ensure an unmixed row and column. Apply Lemma 2.1 to the indicator of A , then use $|R_A| \leq d_2(A)$ and $|Q_A| \leq d_1(A)$. $\square$

We will also use

$$d_i(H \setminus A) = d_i(A). \tag{2}$$

On each cyclic line the number of $1 \rightarrow 0$ transitions equals the number of $0 \rightarrow 1$ transitions, since every entry into a run must be followed by an exit. Summing over the lines proves (2). This symmetry explains why the smaller of a shape and its complement, rather than just the center size, controls the region.

3 The exact two-direction allocation region

Definition 3.1 (Compatible star masks). Three subsets $A, B, C \subseteq H$ are compatible when

$$B \cap (A \cup (A + e_1)) = \emptyset, \quad C \cap (A \cup (A + e_2)) = \emptyset. \quad (3)$$

There is no constraint between B and C . They occupy different leaves, not a common physical fiber.

Theorem 3.2 (Exact small-slack region). *Let $a, b, c \in \mathbb{N}$ satisfy $a + b \leq 169$, $a + c \leq 169$, and put*

$$u = 169 - a - b < 13, \quad v = 169 - a - c < 13.$$

There are compatible masks with exact center size $|A| = a$ and leaf sizes at least $|B| \geq b$, $|C| \geq c$ if and only if

$$\min(a, 169 - a) \leq uv. \quad (4)$$

Proof. Necessity follows from the two disjointness conditions:

$$a + d_1(A) + b \leq 169, \quad a + d_2(A) + c \leq 169.$$

Hence $d_1(A) \leq u$, $d_2(A) \leq v$. The strict upper bounds on u, v allow Proposition 2.2, giving $\min(a, 169 - a) \leq d_1(A)d_2(A) \leq uv$.

For sufficiency, first suppose $a \leq uv$. Inside the nonwrapping rectangle $\{0, \dots, v - 1\} \times \{0, \dots, u - 1\}$ choose the initial row-major Ferrers diagram

$$F_{v,u}(a) = \{(x, y) : 0 \leq x < v, 0 \leq y < u, xu + y < a\}. \quad (5)$$

It has exactly a cells. In each occupied column its first-coordinate section is an initial interval of length less than 13, and in each occupied row its second-coordinate section is likewise an initial interval. Each nonempty section has one exit. Thus $d_1(F_{v,u}(a)) \leq u$ and $d_2(F_{v,u}(a)) \leq v$. If instead $169 - a \leq uv$, take $A = H \setminus F_{v,u}(169 - a)$; equation (2) gives the same bounds and $|A| = a$.

In either case choose

$$B = H \setminus (A \cup (A + e_1)), \quad C = H \setminus (A \cup (A + e_2)). \quad (6)$$

They are compatible and have sizes $169 - a - d_1(A) \geq b$ and $169 - a - d_2(A) \geq c$. If $u = 0$ or $v = 0$, condition (4) forces $a = 0$ or $a = 169$; the empty diagram or its full complement gives the same conclusion, so no division by a zero rectangle width is needed. $\square$

The theorem compares joint feasibility with scalar edge budgets. For $(a, b, c) = (81, 81, 81)$ both pair sums are $162 \leq 169$, but $u = v = 7$ and $\min(81, 88) = 81 > 49$. The requirements therefore cannot be realized. This is a concrete obstruction even though each pairwise budget has slack. The use of lower bounds on the leaves is intentional: the construction takes the full dilation complements. A theorem imposing exact leaf sizes would be a differently phrased statement, not the formal interface used here.

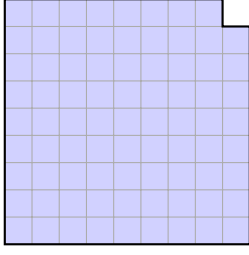
Corollary 3.3 (Sharp balanced occupancy). *Every compatible triple has $\min(|A|, |B|, |C|) \leq 80$. Equality is attained.*

Proof. If all three sizes are at least 81, then $81 \leq a \leq 88$ and the dilation budgets give $d_1(A), d_2(A) \leq 7$. Equation (1) would force $81 \leq 49$, a contradiction. For equality, take

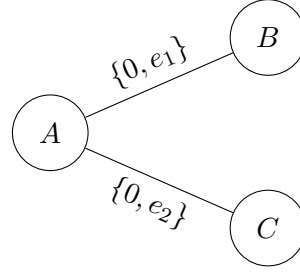
$$A = \{0, \dots, 8\}^2 \setminus \{(8, 8)\}$$

and choose B, C by (6). Here $|A| = 80$, each of its nine occupied rows and nine occupied columns contributes exactly one exit, and $|B| = |C| = 169 - 80 - 9 = 80$. $\square$

$$|A| = 80, d_1 = d_2 = 9$$



A: 9×9 minus one corner



No leaf-leaf conflict edge

Figure 1: An attaining center shape and the normalized two-voltage star. The missing corner reduces 81 to 80 while preserving nine exits in each direction.

Remark 3.4 (General cyclic size; mathematical proof only). The proofs of Lemma 2.1 and Theorem 3.2 work with 169, 13 replaced by p^2, p on $(\mathbb{Z}/p\mathbb{Z})^2$. For odd $p \geq 3$, the largest common guaranteed occupancy is consequently the largest integer k with

$$2k + \lceil \sqrt{k} \rceil \leq p^2.$$

Indeed, writing $t = p^2 - 2k$, a compatible triple with sizes at least k has $k \leq |A| \leq p^2 - k$. If $t < p$, the directional minority bound gives $k \leq t^2$; if $t \geq p$, the same inequality follows from $k \leq p^2$. Thus $\lceil \sqrt{k} \rceil \leq t$ is necessary. For sufficiency use a k -cell Ferrers diagram in a square of side $\lceil \sqrt{k} \rceil$, then the two dilation complements. For admissible k this square has side less than p and each leaf has at least k points. The cyclic and Ferrers Lean modules in this release specialize to $p = 13$; this general- p observation is a hand-proved corollary, not a claim of full parameterized formalization.

4 An actual three-fiber realization in $C_{13}^{\boxtimes 6}$

The mask constraints are not a hypothetical separation profile. We now identify the actual finite graph on which they occur. All coordinates in this section are modulo 13. Define three maps $\phi_i : H \rightarrow H^3$, viewing H^3 as six cycle coordinates, by

$$\begin{aligned} \phi_0(x, y) &= (x, 2x, y, -2y, 0, x), \\ \phi_1(x, y) &= (x - 1, 2x - 1, y, -2y, 0, x - 1), \\ \phi_2(x, y) &= (x - 1, 2x - 1, y, -2y + 1, 0, x + 1). \end{aligned} \tag{7}$$

These are three gauge-shifted BPZ kernel fibers: the kernel is $h(x, y) = (x, 2x, y, -2y, 0, x)$, and the syndrome labels in the upstream encoding are 0, 2197, 2368. Both leaf parameters have been translated by $(-1, 0)$ so that the directional constraints take the form (3).

Lemma 4.1 (Exact physical conflict relation). *The map $(i, z) \mapsto \phi_i(z)$ is injective on $\{0, 1, 2\} \times H$. Each fiber is independent. Between different fibers, the only possible closed-conflict pairs are*

$$\begin{aligned} \phi_0(z) \simeq \phi_1(w) &\iff z = w \text{ or } z = w - e_1, \\ \phi_0(z) \simeq \phi_2(w) &\iff z = w \text{ or } z = w - e_2. \end{aligned}$$

There are no closed conflicts between the two leaf fibers.

Proof. For residues $\delta \in \mathbb{Z}/13\mathbb{Z}$, the simultaneous requirements $\delta, 2\delta \in \{0, 1, -1\}$ imply $\delta = 0$. Within one fiber, the first two coordinates therefore force equal x , and the third and fourth force equal y .

For a center–first-leaf pair, put $\delta = x - s$ and $\eta = y - t$. The relevant differences are $\delta + 1, 2\delta + 1, \eta, -2\eta, 0, \delta + 1$. They all lie in $\{0, 1, -1\}$ exactly when $\eta = 0$ and $\delta \in \{-1, 0\}$. For a center–second-leaf pair the sixth difference is $\delta - 1$, which, together with the first two, forces $\delta = 0$; the third and fourth are $\eta, -2\eta - 1$, forcing $\eta \in \{-1, 0\}$. This proves the two displayed relations.

For a first-leaf–second-leaf pair, the first two differences are $\delta, 2\delta$, so closed conflict would force $\delta = 0$; the sixth difference is then -2 , which is not confusable with zero. Finally the invariants “second minus twice first” distinguish the center from the leaves, and “fourth plus twice third” distinguish the leaves from one another. Along with injectivity within each fiber, these establish injectivity of the full 507-point map. $\square$

For masks define the physical set

$$\mathcal{C}(A, B, C) = \phi_0(A) \cup \phi_1(B) \cup \phi_2(C).$$

Theorem 4.2 (Graph form of the allocation region). *Under all the hypotheses of Theorem 3.2, there are masks whose physical union is independent, whose center fiber has exactly a points and whose leaf fibers have at least b, c points, if and only if $\min(a, 169 - a) \leq uv$.*

Proof. By Lemma 4.1, independence of the physical union is equivalent to (3). Injectivity identifies each physical fiber cardinality with its mask cardinality. Apply Theorem 3.2 in both directions. $\square$

Corollary 3.3 thus gives an actual 240-point independent set in $C_{13}^{\boxtimes 6}$. This realizes a balanced occupancy constraint, not a capacity improvement: the two *complete* leaves already form an independent set of size 338, and the retained 522-dimensional construction has a much larger normalized root. The utility of the theorem is its exact answer when a surrounding problem prescribes joint local occupancies. An insertion into a larger fiber code would still require separation from all outside fibers. No such global gluing or resulting gain is established here.

5 Contextual capacity certificates

We include two previously completed actual-code certificates. This section describes how their sets are constructed and how their exact roots compare to the fixed prior constructions. Their finite recipes are in the companion, and their formal endpoints are imported unchanged.

5.1 The 300-dimensional C_7 construction

A private-pair gadget consists of an independent main code I , selected pairs (r_j, q_j) with all endpoints distinct, $r_j \in I$, $q_j \notin I$ and $\mathcal{N}(q_j) \cap I = \{r_j\}$, two independent complementary ordered transversals P^H, P^V , and an independent auxiliary set X avoiding their *common* neighborhood. Let $R = \{r_j\}$ and $B = I \setminus R$. The auxiliary partition is

$$X^0 = X \setminus (\mathcal{N}(P^H) \cup \mathcal{N}(P^V)), \quad X^H = X \cap \mathcal{N}(P^H), \quad X^V = X \cap \mathcal{N}(P^V).$$

Its profile is $(a, t, s, o, h, v) = (|I|, |R|, |X|, |X^0|, |X^H|, |X^V|)$. The five-dimensional BPZ leaf has profile $(367, 8, 367, 322, 26, 19)$.

In the ordinary paper routing, the Gao product’s main code is the union of seven separated blocks

$$\begin{aligned} I' = & B_L \times B_R \cup R_L \times X_R^0 \cup P_L^H \times X_R^H \cup P_L^V \times X_R^V \\ & \cup X_L^0 \times R_R \cup X_L^H \times P_R^V \cup X_L^V \times P_R^H. \end{aligned}$$

Its private endpoints and ordered transversals are obtained by extending left private pairs over X_R^0 and right private pairs over X_L^0 . For example, $P^H = (P_L^H \times X_R^0) \cup (X_L^0 \times P_R^V)$. The ordinary auxiliary set is $X_L \times X_R$. The upstream Lean routing is the operation `liftRPS S (flip T)` for this paper’s $\text{Gao}(S, T)$; the right flip is essential to the identification.

Tandon’s heterogeneous step keeps that main code, private-pair data and ordered transversals but replaces the auxiliary set by

$$X' = (J_0 \times X_R^0) \dot{\cup} (J_H \times X_R^H) \dot{\cup} (J_V \times X_R^V). \quad (8)$$

Here each J is independent and J_0 avoids the common neighborhood of the left transversals. Right-side class separation makes the three blocks independent and mutually separated. If $q_H = |J_H \setminus \mathcal{N}(X_L^0)|$, $q_V = |J_V \setminus \mathcal{N}(X_L^0)|$ and (s_0, o_0, h_0, v_0) describes J_0 relative to the actual left transversals, then

$$\begin{aligned} s' &= s_0 o_R + |J_H| h_R + |J_V| v_R, \\ o' &= o_0 o_R + q_H h_R + q_V v_R, \\ h' &= h_0 o_R + (|J_V| - q_V) v_R, \\ v' &= v_0 o_R + (|J_H| - q_H) h_R. \end{aligned} \quad (9)$$

The main-code and pair counts remain $a' = (a_L - t_L)(a_R - t_R) + t_L s_R + s_L t_R$ and $t' = t_L o_R + o_L t_R$. Thus an unchanged current main code can still carry a more useful downstream auxiliary geometry.

For the same ordered inputs, ordinary and heterogeneous products are actual siblings: their I , ports and ordered endpoint maps are identical, not merely equicardinal. This justifies using one sibling’s auxiliary set as another’s J_0 . It does *not* identify their neutral sets. In (9), “outside” always means outside the *closed neighborhood of the host’s* X_L^0 , never outside just its endpoint set.

The accepted finite side counts used by this construction are:

	Side set	Neutral reference	Side size	Outside count
C1	J^+	$G_{10}.X^0$	134753	27488
C2	J_{15}	$G_{15}^X.X^0$	49495055	12839823
C3	$G_{15}^{\text{het}}.X$	$G_{15}^X.X^0$	49433743	14045805
C4	J_{30}^{++}	$L_{30}.X^0$	2455726444728097	841760069965664

Here J^+ is an automorphic ten-dimensional code with eight same-size exchanges, J_{15} is an automorphic ordinary fifteen-dimensional code, and J_{30}^{++} is an automorphic code obtained from two G_{15}^{het} inputs. All their literal data and identities are included in the Lean dependencies; these numbers are not assumed profiles. In particular C4 refers to the ordinary L_{30} , not its modified auxiliary sibling.

For completeness the compressed counting principle is as follows. Write a source as a disjoint union of products and a reference as a union of products. For each source factor, record a Boolean vector of membership in the closed neighborhoods of all reference factors, together with the exact frequency of each vector. On concatenating coordinates take componentwise conjunction and multiply frequencies; sum frequencies that yield the same vector. The all-false final vector counts exactly the points outside the reference neighborhood, since neighborhoods of products factor coordinatewise. Summing over the disjoint source blocks gives the whole count. The Lean implementation links its finite lookup tables and bitset operations to this actual graph predicate.

The seven-family realization associated to a gadget is

$$\Phi(G) = (B, N, A, D, O, H, V) = (I \setminus R, X^0, X^V, X^H, R, P^H, P^V).$$

Each family is independent and the required separated pairs are

$$BO, BH, BV, NA, ND, NO, NH, NV, AD, AH, DV. \quad (10)$$

When literal label words have a separating pair in some coordinate, their physical product blocks are independent and disjoint. Their union has exact size

$$W_K(w_1, \dots, w_r) = \sum_{z \in K} \prod_{i=1}^r w_i(z_i). \quad (11)$$

This is the BPZ layered substitution principle, not a numeric-only profile propagation rule.

The retained strong construction makes three auxiliary refinements, at dimensions 15, 30 and 40, using genuine sibling identities. The resulting forty-dimensional gadget has seven-family weights

$$\begin{aligned} &(305594401588752181313, 175215226049998841399, \\ &89250516720400951188, 65106977219472028790, \\ &26158021392280384832, 26158021392280384832, 26158021392280384832). \end{aligned}$$

Use this realization and two copies of the inherited flipped fifty-five-dimensional realization in BPZ's S_{3b} substitution. The output has dimension 150. Apply to two copies the terminal code

$$K_2 = \{BB, HN, NH, HD, VA, DV, AH\}.$$

Its seven words are pairwise separated under (10). For an identical input vector $(b, n, a, d, 0, t, t)$ its weight is $b^2 + 2t(n + a + d)$. Equation (11) therefore gives an actual 300-dimensional code of size M_{300} , recorded in Appendix B. The full dependency recipe, not just the top weights, accompanies the source.

Theorem 5.1 (Retained C_7 certificate). *There is a code in $C_7^{\mathbb{R}^{300}}$ of size M_{300} , and*

$$3258834805519757^{300} < M_{300}(10^{15})^{300}.$$

Consequently $\Theta(C_7) > 3.258834805519757$.

Proof. The actual gadget and sibling identities establish the input realizations; the certified S_{3b} and K_2 tables establish independence and the exact weighted cardinality. The finite integer inequality is part of the accepted Lean certificate. Taking positive 300th roots and using the defining capacity lower bound proves the strict endpoint. $\square$

The companion also retains the 280-dimensional simple terminal size M_{280} . Its selected prior 560-dimensional Stavriianov certificate has size $M_S = M_{280}^2$; this is an exact identity between the listed integers, not an equality of capacities. The accepted comparison $M_{280}^{300} < M_{300}^{280}$ gives

$$M_S^{15} < M_{300}^{28}, \quad M_S^{1/560} < M_{300}^{1/300}.$$

The earlier Tandon 500-dimensional root is also exceeded, but Table 1 uses the stronger chosen prior root.

5.2 The guarded 522-dimensional C_{13} construction

The construction uses Protti's 58 typed actual base cells, each formed from explicit BPZ syndrome fibers. Their independence, required separations and cardinalities are proved using the actual sets **BaseC13.Pre**. They are not formal letters whose sizes are postulated. Long's 49-node recursion DAG, input orientations and K_{3a} terminal are retained. The only additions are the following guarded reference words, with child order part of the definition:

Node	Ordered children	Output	References	Added typed cells
n_{12}	n_6, n_{10}, n_{11}	A	BAB, BBV	6
n_{16}	n_{11}, n_{14}, n_{15}	A	BAB, BBV	3
n_{36}	n_{35}, n_9, n_{21}	A	BAB, BBV	3
n_{17}	n_5, n_{12}, n_{16}	N	$BBA, BBH, BVB, VAB, VBH, VHB$	15
n_{37}	n_{20}, n_{34}, n_{36}	N	same six words	21

The final typed tables contain 4460 generator words and 300 terminal words. The terminal children are n_{17}, n_{37}, n_{48} , of block dimensions 34, 36, 17: in actual cycle coordinates the dimension is $6(34 + 36 + 17) = 522$.

A type records the required avoidance relations with output families. After a child changes, masks are recomputed for that actual child context. The chosen extension cells avoid the output and its required neighbors, while retained cells acquire conservative mask bits. Literal rows certify internal independence, required cross-row separation and absence of duplicate generator words. These statements build actual `Realisation.multiSubst` objects. Their cardinality identities come from (11); repeated occurrences of a shared DAG child use *new coordinate blocks*, so sharing is not physical coordinate reuse. The flattening graph isomorphism gives a finite code in the actual cycle strong power.

Joint additions need not have monotone downstream gains: an enlarged child avoidance mask can shrink later usable cells. This is why the accepted final tables are those for the complete chosen combination, not independent increases of scalar intermediate profiles.

Theorem 5.2 (Retained C_{13} certificate). *There is an independent set of size N_{522} in $C_{13}^{\boxtimes 522}$. If M_L is Long’s fixed 522-dimensional certificate size, then*

$$N_{522} > M_L, \quad 6302927403571109^{522} < N_{522}(10^{15})^{522}.$$

In particular $N_{522}^{1/522} > M_L^{1/522}$ and $\Theta(C_{13}) > 6.302927403571109$.

Proof. The typed base realization, checked substitutions and terminal code construct the actual independent set. Their proved weighted identities give its cardinality N_{522} . The two exact integer comparisons are in the accepted capacity and strict-display modules. Monotonicity of positive 522th roots and the code-to-capacity inequality prove both claims. Both full integers appear in Appendix B. $\square$

The new local region is logically separate from this certificate: the masks of Sections 2–4 have not been glued into its full DAG or base fibers. Neither strict endpoint changed in the structural round.

6 Formal statements, trust and reproduction

The project pins Lean 4.32.2 and mathlib revision

905b95818eb32af7874a58b427f50c1711a5e96c.

Only selected transitive local imports are bundled. BPZ, Long and Protti source headers and Apache-2.0 licensing are retained; mathlib is an external dependency. The following map identifies the endpoints used by the mathematical claims:

Claim	Module and declaration
Minority rectangle	MinorityRectangle: minority_rectangle
Exact abstract region	StarRegion: star_region_iff
Actual graph region	ActualRegion: actual_star_region_iff
Balanced optimum and 240 points	ActualRegion: actual_star_upper80, actual_star_attains80; PhysicalStar: cycleCode_independent, cycleCode_card
C_7 strict endpoint	C7Improvement.Capacity: shannonCapacity_C7_gt_strong
C_{13} strict endpoint	ExplorationC13FrontierR03.StrictDisplay: strict_capacity_lower
Classical C_7 illustration	ExplorationC7.Correlated.Geometry, ExplorationC7.Correlated.Witness

The new aliases `StructuralRelease.c7_strict`, `c13_strict`, `c13_region`, `c13_example` and `c7_illustration` preserve these statement types. In particular `c13_region` retains both pair budgets, both strict slack hypotheses, exact center size and lower-bound leaf sizes.

The minority-rectangle argument is symbolic and uses standard logic. The cyclic and Ferrers templates, literal construction tables, and physical relation use `native_decide`. The physical check has 507 points. This adds compiler trust; it is not advertised as purely kernel arithmetic. The accepted audits report 62 native dependencies in the selected strong C_7 strict closure; 166 in the selected C_{13} endpoint union including strict display (161 new, 5 upstream); 9 in the eleven selected local structural declarations (6 new, 3 BPZ); and 8 in twenty selected C_7 illustration closures. The six-declaration historical C_7 audit union has 66, which is a different count from the single endpoint's 62. Otherwise the reported axioms are standard `propext`, `Classical.choice` and `Quot.sound`; the selected accepted audits contain no holes or unexpected mathematical axioms.

The structural necessity proof ranges over *arbitrary* masks. Only the finite sufficiency templates are checked computationally: `ferrers_statistics` verifies all $13 \cdot 13 \cdot 170$ parameter triples satisfying the rectangle-size condition, including complement templates. The physical relation check covers the 507-point three-fiber map, after which arbitrary compatible masks are handled symbolically. This separates a finite certificate from its semantic transport rather than assuming candidate densities.

For this publication we copied exact-source accepted compiled artifacts locally and compiled *only the new aggregate integration target*. That validates imported statement types and coexistence of the selected namespaces; it is not a fresh clean-clone execution of the original large finite checks. Accepted audit reports are included as provenance, not regenerated here. No old side counts, baseline audits or searches were rerun. A reader's clean build executes those original finite certificates from source:

```
cd lean
lake exe cache get
lake build StructuralRelease
```

The README gives prerequisites, the logged reproduction wrapper and the scope of the performed validation. Caches and compiled files are not part of the release. The public repository is

<https://github.com/JingzheShi/odd-cycle-shannon-structural-lean>.

A Classical correlated separation in C_7^3

For a complete rectangle $\mathcal{R}(P) = \prod_i P_i$ one has

$$\mathcal{R}(P) \perp \mathcal{R}(Q) \iff \text{there is a coordinate } i \text{ with } P_i \perp Q_i. \quad (12)$$

The reverse implication is immediate. For the other direction, if each coordinate fails separation, choose one confusable pair there; their concatenations are a confusable pair of rectangle points. This proof also handles empty factors, whose separation is vacuous. For arbitrary correlated sets the exact criterion instead is

$$S \perp T \iff \forall x \in S \forall y \in T \exists i : x_i \not\sim y_i. \quad (13)$$

The separating coordinate may depend on the pair. These are classical strong-product and graph-packing principles; complement-homomorphism and packing treatments give the wider context [10, 11].

Here is a literal, nonaffine example:

$$\begin{aligned} P &= \{(0, 0, 0), (1, 2, 1), (2, 4, 2), (3, 6, 3), \\ &\quad (4, 1, 4), (5, 5, 5), (6, 3, 6)\}, \\ Q &= \{(0, 0, 2), (1, 1, 4), (2, 2, 6), (3, 4, 0), \\ &\quad (4, 3, 5), (5, 6, 1), (6, 5, 3)\}. \end{aligned}$$

Both sets are independent, $P \perp Q$, and each coordinate image of each set is all of $\mathbb{Z}/7\mathbb{Z}$. Thus their rectangular hulls both equal the entire 343-point ambient space, so no universal separating coordinate exists. Their second-coordinate permutations are not of the form $x \mapsto ax + b$: in P the entries at 0, 1 force $b = 0, a = 2$, inconsistent with the entry at 5; in Q they force $b = 0, a = 1$, inconsistent with the entry at 3.

Each of the three coordinates is essential for the stated P/Q separation, as the following pairs are nonconfusable in exactly the listed coordinate:

Only separating coordinate	Point of P	Point of Q
0	(0, 0, 0)	(5, 6, 1)
1	(1, 2, 1)	(0, 0, 2)
2	(0, 0, 0)	(0, 0, 2)

Deleting any coordinate makes the corresponding pair confusable. More generally a projection obtained only by deleting or repeating coordinates can preserve this separation only if its coordinate map is surjective onto all three original coordinates. This does not restrict arbitrary recoding of the points.

The two whole blocks already fit BPZ’s arbitrary **Realisation** interface as two separated virtual letters; no new composition machinery is needed. The literal witness and its graph claims are formalized, but its 14 points are fewer than the simple product’s $3^3 = 27$. One completed fixed-context test on the 78 points outside the closed neighborhood of $P \cup Q$ explored 4652 search nodes and retained at best 27 points. That recorded result concerns only this particular completion problem, not every correlated code and not $\Theta(C_7)$. No capacity gain or new lower-bound mechanism is claimed for the illustration.

B Exact certificate integers

The following displays denote decimal concatenation of their lines; line breaks are not multiplication. These integers also appear as literal definitions in the corresponding Lean modules and as decimal strings in `data/exact_roots.json`. M_{280} identifies the chosen Stavriianov integer through $M_S = M_{280}^2$.

$$M_{280} = \left[\begin{array}{l} 454402134191994507126819903596564906706672995987342041247384 \\ 728314329074657273274845475279319219681718758139567086979352 \\ 344706380862194486578561 \end{array} \right]_{\text{concat}}.$$

$$\begin{aligned}
M_{300} &= \left[\begin{array}{c} 829285028795068306996156269127051491792580794837082180052404 \\ 457654405642151270168458082344281085433541502968490704472521 \\ 4369985043992527534955657521950273 \end{array} \right]_{\text{concat}} . \\
N_{522} &= \left[\begin{array}{c} 229658862260924106595253952272045722516666849084998891003074 \\ 56594720843747777984843267498257757027599220802987164115637 \\ 395623399737872969052176585212885966843594415072366106347620 \\ 633527919020308823654200143088901344335953552868018939685799 \\ 436735561005316491462893882696633487887255686360035537551619 \\ 671849877807457341677220644469448879804632334817769471949487 \\ 8137521970969100483646258158547494769299673652709061296128 \end{array} \right]_{\text{concat}} . \\
M_L &= \left[\begin{array}{c} 229652548049794108669807110452510507772003551488435507257929 \\ 181011771863281382028370787610380986376743189829161636850800 \\ 135155371587104528625458030017679021783209914609549343731326 \\ 915703357857186485927010917180050856575156133657818150496082 \\ 954825607329536419413840822416591512368164009868691426795406 \\ 198331457870964794955825596500475143546966222926330319233036 \\ 5907306254086949021017927545131180608697270602131050594304 \end{array} \right]_{\text{concat}} .
\end{aligned}$$

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